Portfolio Project Data Analytics

TrendOne Customer Analytics Report (Revised)

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Python, SQL (PostgreSQL), Power BI.

https://github.com/nigelalessan/trendone_customer_behavior_analysis

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1. Executive Summary

Overview

This project explores customer shopping behavior to identify patterns in purchase amounts, evaluate relationships between customer characteristics and purchasing behavior, and develop exploratory customer segments.

The analysis combines Python-based data preparation and statistical analysis, SQL-based data validation and business queries, K-Means clustering, and a Power BI dashboard. The goal is to demonstrate how customer data can be transformed into structured findings that support business investigation and decision-making.

TrendOne Co. is a simulated business case created for this portfolio project. The analysis does not represent work performed for an actual company.

Key Findings

1. **No meaningful purchase amount differences were detected across the groups examined.**

The statistical tests did not find statistically significant differences in recorded purchase amounts across the analyzed demographic and customer groups, including age groups, gender, region, product category, discount usage, and subscription status. The observed effect sizes were also very small.

2. **Subscription status and discount usage showed a strong association.**

The chi-square test indicated a strong association between subscription status and discount usage ($\chi^2 = 1908.92, \quad p < 0.001, \quad$ Cramér's $V = 0.70$). This relationship deserves further investigation, particularly to determine whether it reflects business rules, promotion eligibility, or characteristics of the dataset.

3. **Six exploratory customer segments were identified.**

K-Means clustering produced six customer profiles based on selected numerical features. However, the silhouette score was 0.2864, indicating limited separation between clusters. The profiles should therefore be treated as exploratory groups rather than validated business segments.

Business Implications

The findings suggest that customer purchase amounts should not be explained using demographic characteristics alone. The strong subscription discount association warrants a closer review of the underlying business rules and data structure. Meanwhile, the exploratory segments provide a starting point for further customer investigation, not a direct basis for targeted campaigns without validation.

The analysis also highlights a data limitation, the dataset does not contain transaction timestamps or detailed transaction histories, preventing a reliable measurement of actual customer retention over time.

4. Business Context & Questions

Business Context

Understanding customer behavior can help businesses evaluate purchasing patterns, investigate customer engagement, and identify opportunities for improving customer experiences.

However, observed patterns do not necessarily indicate the reasons behind customer behavior. A reliable analysis should distinguish descriptive findings, statistical evidence, and business hypotheses that require further investigation.

For this project, a structured customer shopping dataset was analyzed to demonstrate an end-to-end customer analytics workflow.

Project Objectives

The project aims to:

- Understand the structure, quality, and limitations of the available data.
- Examine customer purchasing behavior across selected customer characteristics.
- Apply statistical methods to evaluate relationships and differences between groups.
- Develop exploratory customer segments using K-Means clustering.
- Validate data and analytical outputs using SQL and Python.
- Translate the findings into practical recommendations and identify additional data requirements.

Key Business Questions

Business Area	Analytical Question
Customer Value	Which observed customer characteristics are associated with purchase amount?
Customer Value	Do recorded purchase amounts differ across selected customer groups?
Product Behavior	How does purchase amount vary across product categories?
Promotions	Is discount usage associated with purchasing behavior?
Customer Engagement	Does purchase amount differ between subscribed and non-subscribed customers?

Customer Engagement	How strongly are subscription status and discount usage associated?
Customer Retention	Can customer retention be measured using the available data?
Customer Segmentation	Can exploratory customer profiles be identified from selected numerical features?

The retention question cannot be answered directly because the dataset lacks transaction timestamps and longitudinal customer activity records. Previous Purchases and Frequency of Purchases can provide limited behavioral context, but they are not substitutes for actual retention metrics.

5. Data Understanding & Preparation

Dataset Overview

The dataset contains 3,900 rows and 18 columns describing customer shopping behavior.

The available variables include customer identifiers, demographic characteristics, product categories, purchase amounts, discount usage, subscription status, previous purchases, purchase frequency, and review ratings.

Data overview

Item	Result
Records	3,900
Columns	18
Unique Customer IDs	3,900
Exact duplicate rows	None detected
Missing Review Rating values	37
Missing Review Rating percentage	Approximately 0.95%

The dataset contains one record per customer identifier in the provided file. This structure should not be interpreted as a complete transaction history or a record of all purchases made by each customer.

Dataset source: https://github.com/amlanmohanty1/customer-trends-data-analysis-SQL-Python-PowerBI/blob/main/customer_shopping_behavior.csv

Data Quality Assessment

The initial assessment examined dataset dimensions, column types, identifier uniqueness, exact duplicate records, missing values, and the suitability of variables for analysis.

The main data quality observations were:

- Customer ID was treated as an identifier rather than a numerical measurement.
- No exact duplicate rows were detected.
- Thirty-seven Review Rating values were missing.

- Review Rating missing values were preserved as missing rather than replaced with zero, because zero would represent a different meaning from an unavailable rating.
- Age Group was derived from age to support grouped comparisons and reporting.

Data Preparation Workflow

The preparation process followed these steps:

1. Load and inspect the raw dataset.
2. Review column names, data types, unique values, and missing values.
3. Validate customer identifiers and check exact duplicate records.
4. Apply appropriate data types and prepare analytical variables.
5. Create age groups for demographic comparisons.
6. Export the prepared dataset for subsequent analysis.
7. Use the prepared data as the common basis for statistical analysis, segmentation, SQL queries, and dashboard reporting where applicable.

Data flow: Raw Dataset → Data Quality Assessment → Data Preparation → Exploratory Analysis → Statistical Analysis & Segmentation → SQL Validation and Business Queries → Power BI Dashboard.

The workflow was designed to keep the data preparation process transparent and make the analysis easier to reproduce.

4. Analytical Methodology

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used to understand distributions, compare customer groups, and identify patterns requiring further investigation.

The analysis examined purchase amounts, demographic characteristics, product categories, subscription status, discount usage, previous purchases, purchase frequency, and review ratings.

Descriptive statistics and visualizations provided context for the subsequent statistical tests and segmentation process.

Statistical Analysis

Statistical tests were selected according to the types of variables and the analytical questions.

Analytical Question	Method	Purpose
Association between age and purchase amount	Spearman correlation	Evaluate monotonic association
Association between previous purchases and purchase amount	Spearman correlation	Evaluate monotonic association
Purchase amount between two independent groups	Welch's t-test	Compare group means without assuming equal variances
Purchase amount across more than two groups	Welch's ANOVA	Compare group means without assuming equal variances
Association between two categorical variables	Chi-square test of independence	Evaluate whether categorical variables are associated

The significance threshold was set at ($\alpha = 0.05$) Effect sizes were also considered so that interpretation did not rely solely on p-values.

A statistically significant result indicates evidence of an association or difference under the test's assumptions. It does not, by itself, establish causality or demonstrate business importance.

Customer Segmentation

K-Means clustering was applied to selected numerical customer features to explore whether customers could be grouped into profiles with similar characteristics.

The final model used the following parameters:

- Number of clusters: 6
- Random state: 42
- Number of initializations: 20
- Silhouette score: 0.2864

The silhouette score suggests that separation between clusters is limited. Cluster descriptions are therefore relative to the observed dataset and should not be interpreted as established customer personas, measures of profitability, or evidence of loyalty.

SQL Validation and Analysis

SQL was used to support data validation and customer-level business analysis. Python and SQL results were compared for relevant metrics to check consistency between the analytical workflows.

SQL provides a complementary way to inspect data, calculate customer metrics, and answer structured business questions. Agreement between the two implementations supports consistency but does not independently prove that every assumption or analytical interpretation is correct.

5. Statistical Findings

Summary of Results

The following table summarizes the main statistical results.

Relationship Tested	Result	Interpretation
Age vs. Purchase Amount	Spearman $(\rho = -0.01)$, $(p = 0.51)$	Negligible monotonic association detected
Gender vs. Purchase Amount	Welch's $(t = -0.88)$, $(p = 0.38)$, $(d = -0.030)$	No statistically significant difference detected; negligible effect
Previous Purchases vs. Purchase Amount	Spearman $(\rho = 0.008)$, $(p = 0.604)$	Negligible monotonic association detected
Region vs. Purchase Amount	Welch's $(F = 1.679)$, $(p = 0.169)$, partial $(\eta^2 = 0.001278)$	No statistically significant difference detected
Age Group vs. Purchase Amount	Welch's $(F = 0.462)$, $(p = 0.805)$, partial $(\eta^2 = 0.000585)$	No statistically significant difference detected
Product Category vs. Purchase Amount	Welch's $(F = 1.357)$, $(p = 0.254)$, partial $(\eta^2 = 0.001118)$	No statistically significant difference detected
Discount Usage vs. Purchase Amount	Welch's $(t = -1.112)$, $(p = 0.266)$, $(d = -0.036)$	No statistically significant difference detected
Subscription Status vs. Purchase Amount	Welch's $(t = -0.44)$, $(p = 0.66)$, $(d = -0.016)$	No statistically significant difference detected
Subscription Status vs. Discount Usage	$(\chi^2 = 1908.92)$, $(p < 0.001)$, Cramér's V = 0.700	Strong association detected

Finding 1: Purchase Amount Differences Were Small

The tests did not detect statistically significant purchase amount differences across the examined demographic and customer groups. The reported effect sizes were also very small.

This means the analysis does not provide sufficient evidence that these individual variables distinguish customers by recorded purchase amount in this dataset.

It does not prove that all groups are identical or that these variables are irrelevant in every business context. Additional features, transaction-level data, or a different dataset may reveal other patterns.

Finding 2: Subscription Status and Discount Usage Were Strongly Associated

The chi-square test found a strong association between subscription status and discount usage, with Cramér's V of approximately 0.70.

This is the most prominent categorical relationship identified in the analysis. It should be investigated before making business decisions.

Possible follow-up checks include:

- Reviewing the subscription and discount eligibility rules.
- Inspecting the underlying cross-tabulation and cell counts.
- Checking whether the relationship is partly determined by the dataset's structure or generation process.
- Validating the pattern using transaction-level records from an actual business environment.

The result does not establish that subscribing causes discount usage or that discounts cause customers to subscribe.

Finding 3: Customer Retention Cannot Be Measured Reliably

The dataset does not provide transaction timestamps or a sequence of customer activities over time. Consequently, metrics such as repeat purchase rate over a defined period, customer retention rate, and time between purchases cannot be calculated reliably.

Previous Purchases and Frequency of Purchases can be explored as recorded customer attributes, but they do not establish when purchases occurred or whether customers remained active over a specific period.

6. Exploratory Customer Segmentation

Segment Profiles

K-Means produced six clusters with the following descriptive profiles.

Cluster	Customers	Avg. Purchase Amount	Avg. Previous Purchases	Avg. Review Rating
C0	647	$76.00	12.11	4.34
C1	673	$45.99	38.88	3.15
C2	665	$39.48	11.90	3.31
C3	646	$80.33	39.76	4.22
C4	625	$82.77	20.78	3.06
C5	644	$35.84	28.39	4.43

The cluster labels below are descriptive summaries of the observed profiles, not validated business personas.

Cluster	Descriptive Profile	Potential Follow-up
C0	Higher-spend, lower purchase history	Examine what distinguishes higher recorded purchase amounts among customers with fewer previous purchases
C1	Repeat-history, lower-rating	Investigate the low average rating and review patterns
C2	Lower-spend, lower purchase history	Explore the characteristics associated with this lower-spend profile
C3	Higher-spend, higher purchase history	Examine its product, demographic, and purchasing characteristics
C4	Higher-spend, lower-rating	Investigate possible customer experience or product-related issues
C5	Lower-spend, higher-rating	Explore whether positive ratings coexist with lower recorded purchase amounts

Model Evaluation and Interpretation

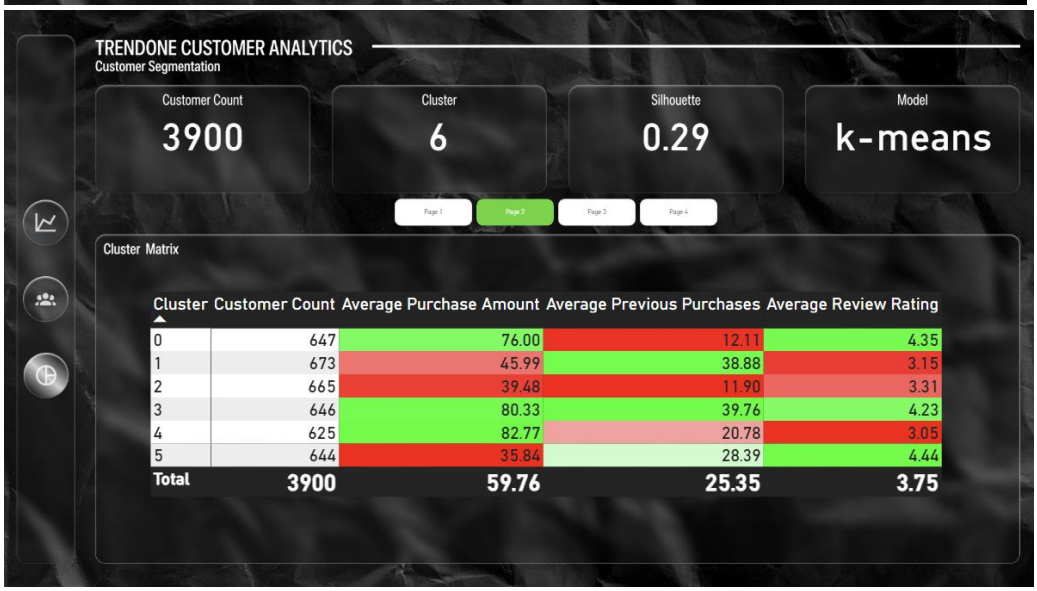
The final silhouette score was 0.2864, indicating limited separation between the clusters. The analysis also examined sensitivity to different random initializations using a separate single-initialization experiment.

That experiment produced a mean Adjusted Rand Index (ARI) of 0.5145 with a standard deviation of 0.2411 across the tested initializations. This indicates that the clustering assignments were sensitive to initialization under that experiment. It is not a resampling stability test and does not directly measure the stability of the final model using 20 initializations.

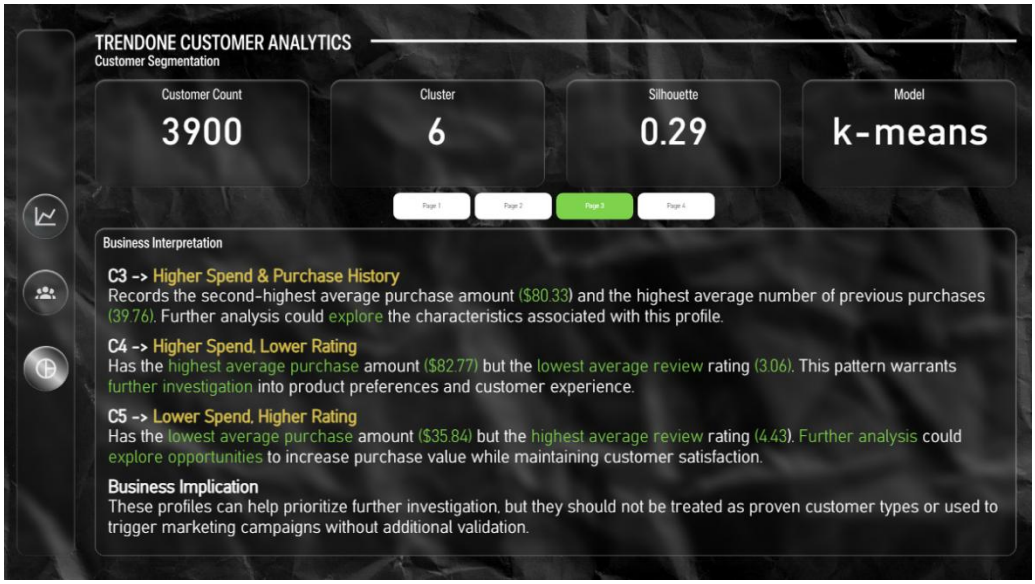
These limitations mean the segments should be used as hypotheses for further analysis rather than immediately deployed for campaign targeting or customer prioritization.

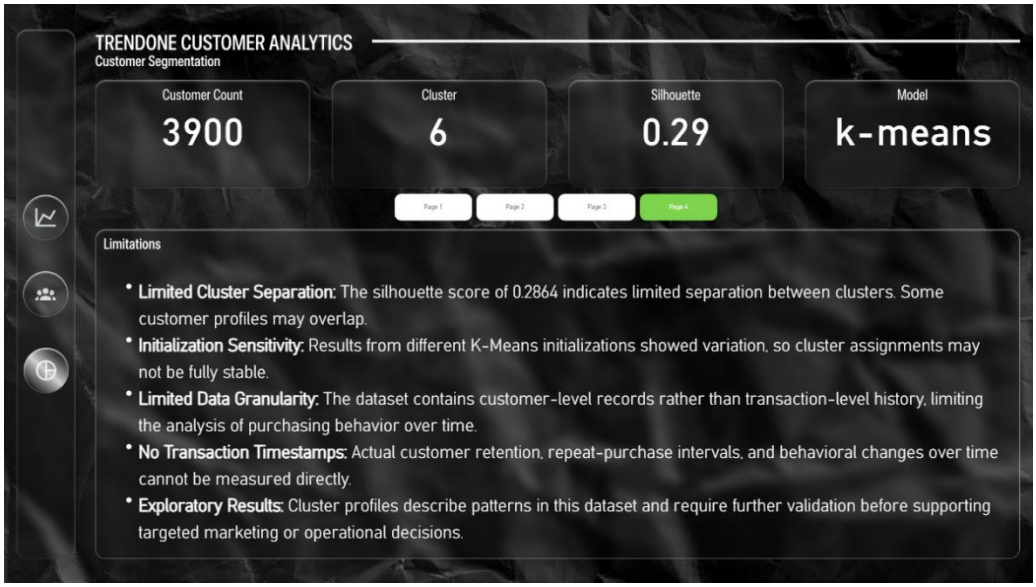
Customer Segmentation Dashboard





Cluster	Customer Count	Average Purchase Amount	Average Previous Purchases	Average Review Rating
0	647	76.00	12.11	4.35
1	673	45.99	38.88	3.15
2	665	39.48	11.90	3.31
3	646	80.33	39.76	4.23
4	625	82.77	20.78	3.05
5	644	35.84	28.39	4.44
Total	**3900**	**59.76**	**25.35**	**3.75**


TRENDONE CUSTOMER ANALYTICS
Customer Segmentation
Customer Count
3900
Cluster
6
Silhouette
0.29
Model
k-means
Business Interpretation
C3 -> Higher Spend & Purchase History
Records the second-highest average purchase amount ($80.33) and the highest average number of previous purchases (39.76). Further analysis could explore the characteristics associated with this profile.
C4 -> Higher Spend, Lower Rating
Has the highest average purchase amount ($82.77) but the lowest average review rating (3.06). This pattern warrants further investigation into product preferences and customer experience.
C5 -> Lower Spend, Higher Rating
Has the lowest average purchase amount ($35.84) but the highest average review rating (4.43). Further analysis could explore opportunities to increase purchase value while maintaining customer satisfaction.
Business Implication
These profiles can help prioritize further investigation, but they should not be treated as proven customer types or used to trigger marketing campaigns without additional validation.


TRENDONE CUSTOMER ANALYTICS
Customer Segmentation
Customer Count
3900
Cluster
6
Silhouette
0.29
Model
k-means
Limitations
• Limited Cluster Separation: The silhouette score of 0.2864 indicates limited separation between clusters. Some customer profiles may overlap.
• Initialization Sensitivity: Results from different K-Means initializations showed variation, so cluster assignments may not be fully stable.
• Limited Data Granularity: The dataset contains customer-level records rather than transaction-level history, limiting the analysis of purchasing behavior over time.
• No Transaction Timestamps: Actual customer retention, repeat-purchase intervals, and behavioral changes over time cannot be measured directly.
• Exploratory Results: Cluster profiles describe patterns in this dataset and require further validation before supporting targeted marketing or operational decisions.

7. Recommendations & Limitations

Recommendations

1. Investigate the Subscription–Discount Relationship

Review the relationship between subscription status and discount usage at the data and business-rule levels. Determine whether discount access is linked to subscription eligibility or whether the observed association reflects a data-generation constraint.

Before evaluating the effectiveness of subscription programs or discounts, use transaction-level records and a design capable of distinguishing association from causal effects.

2. Avoid Using a Single Variable as a Measure of Customer Value

The analysis did not find meaningful purchase amount differences across the groups examined. Customer value should therefore not be inferred from demographic characteristics, subscription status, or previous purchase count alone.

A more complete assessment would require historical net spending, purchase frequency over a defined period, profit margin, returns, and customer activity over time.

3. Validate the Exploratory Customer Segments

Review the cluster profiles with additional features and, where possible, compare them against real business outcomes. Test whether the profiles are stable, interpretable, and useful for a specific decision.

Do not launch targeted campaigns solely because a cluster has a descriptive label. Any proposed intervention should be evaluated using suitable business metrics and an appropriate test design.

4. Improve the Data Foundation

For stronger customer analytics, collect or make available:

- Transaction ID and transaction timestamp.
- Stable customer identifiers across transactions.
- Product-level quantity, price, discount, and net sales.
- Promotion identifiers and discount eligibility rules.
- Product cost or margin data.
- Customer acquisition and activity history.
- Campaign exposure and response records.

- Consistent observation windows for retention analysis.

These improvements would support more reliable customer value measurement, cohort analysis, retention reporting, and campaign evaluation.

Limitations

The results should be interpreted within the following constraints:

- **Dataset scope:** Findings describe the provided dataset and should not be generalized automatically to other customer populations.
- **Data granularity:** The file contains one record per customer identifier rather than a detailed transaction history.
- **Missing values:** Thirty-seven Review Rating values were missing and were retained as missing.
- **No timestamps:** Actual retention, purchase intervals, and time-based customer activity could not be measured.
- **Statistical interpretation:** Statistical tests do not establish causality. Non-significant results do not prove that no relationship exists.
- **Segmentation quality:** The silhouette score indicates limited cluster separation, and initialization sensitivity warrants caution.
- **Business validation:** The clusters and recommendations have not been validated through real business interventions or operational outcomes.

Conclusion

This project demonstrates an end-to-end customer analytics workflow, from data quality assessment and preparation to statistical testing, SQL analysis, exploratory segmentation, and dashboard reporting.

The main value of the analysis is not simply identifying patterns, but distinguishing findings supported by the available data from questions that require additional evidence.

The next step is to validate the subscription-discount relationship, improve the availability of transaction-level data, and assess whether the exploratory customer profiles offer measurable value in a real business setting.

Appendix: Reproducibility

Tools and Methods

- **Python:** Data preparation, exploratory analysis, statistical testing, and K-Means clustering.
- **SQL / PostgreSQL:** Data validation, customer metrics, and business queries.
- **Power BI:** Interactive reporting and customer behavior visualization.
- **Statistical methods:** Spearman correlation, Welch's t-test, Welch's ANOVA, chi-square test of independence, effect sizes, and silhouette analysis.

Repository Structure

The repository contains the raw and prepared datasets, analysis notebooks, SQL scripts, Power BI dashboard, documentation, architecture diagrams, and analytical outputs.

Repository: https://github.com/nigelalessan/trendone_customer_behavior_analysis

Reproduction Steps

1. Download or access the dataset from the documented source.
2. Review the data requirements and preparation documentation.
3. Run the notebooks in sequence, following their documented dependencies.
4. Execute the SQL scripts against the prepared dataset using the documented database setup.
5. Review the statistical outputs and clustering evaluation.
6. Open the Power BI file and configure the data source path if necessary.

Environment and dependency versions: python 3.12, postgresql 18.4, jupyter notebook 7.2.2, pandas 3.0.6, matplotlib 3.11.2, seaborn 0.13.2, scipy 1.18.1, pingouin 0.7.0, scikit-learn 1.9.1, Power BI 2.159.883.0

Dataset source and license: https://github.com/amlanmohanty1/customer-trends-data-analysis-SQL-Python-PowerBI/blob/main/customer_shopping_behavior.csv

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This report was prepared as an independent portfolio project using a simulated business case. It does not represent client work or verified outcomes from a real company.